

International Coordination of Trade and Green Industrial Policies*

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Abstract

Green industrial policy is often justified as a way to loosen domestic political constraints on carbon mitigation. Yet politically effective green subsidies must discriminate in favor of domestic producers. These subsidies can have negative international spillovers, as they may lead to production relocation and profit shifting. In this paper I argue that green subsidies can also have negative *political* spillovers, as they make it more difficult for foreign policymakers to implement climate policy. To study this logic I develop a two-country model in which producers can adopt a dirty or green technology and the political weight of a sector scales with the number of domestic firms. In the noncooperative equilibrium, carbon taxes are too low and one of two outcomes obtains: a subsidy race, or international specialization, in which one country implements both subsidies and mitigation while the other remains stuck in a no-mitigation equilibrium. The optimal trade-climate agreement commits countries to free trade, sets a globally efficient carbon price, and allows but caps discriminatory green subsidies when domestic constraints bind. The results imply that post-WTO rules should coordinate (rather than prohibit) green subsidies: they can be instrumentally valuable for mitigation, but their uncoordinated use generates beggar-thy-neighbor political effects.

1. Introduction

Green industrial policies have emerged in part from the need to relax political constraints to more stringent climate policies (Meckling *et al.*, 2015, 2017; Juhász and Lane, 2024). This political logic requires that policies supporting green industries be targeted to politically pivotal domestic

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producers. For this reason, green industrial policies are public goods in the international arena (because they lead to reductions in greenhouse gas (GHG) emissions) but, crucially, also have a zero-sum component, as they shift investment and profits from foreign countries to the domestic economy (Allcott *et al.*, 2024). The zero-sum distributive implications of green industrial policies at the international level can lead to “subsidy races” (Clausing and Wolfram, 2023; The Economist Intelligence Unit, 2023). However, we lack a theoretical understanding of the welfare and political implications of this logic: will subsidy races accelerate the green transition and increase global welfare, or will competition on subsidies lead to inefficiencies, waste of fiscal resources, and potentially a political backlash to green policies?

Domestic content requirements in green industrial policies such as the now defunct Inflation Reduction Act of 2022 violate World Trade Organization rules (Charnovitz and Fischer, 2015; World Trade Organization, 2024). This raises a second question: how would a post-WTO world deal with the increase in green subsidies across both developed and developing countries? We can speculate that new forms of international cooperation could emerge to ameliorate the zero-sum effects of green subsidies. For example, countries that want to keep free trade flows and at the same time support domestic green industries could coordinate their subsidies on different parts of critical supply chains.

To answer these questions, I develop a framework that integrates a theory of domestic climate policymaking into a model of international trade and climate cooperation. The economic model is based on the partial-equilibrium version of Venables (1987) (revisited by Ossa, 2011) presented in Helpman and Krugman (1989, Ch. 7), and revisited by Bagwell and Staiger (2015). The model features two economies with two sectors, one of which has monopolistic competition with increasing returns to scale and trade costs. The main feature of this model is the “firm delocation” effect, by which tariffs induce foreign firms to relocate in the home economy, which allows home consumers to save on transport costs. I introduce two modifications. First, I introduce two technologies into the monopolistically competitive sector, one of which releases carbon emissions and thus creates a global negative externality. This allows me to study climate policy, particularly carbon taxes and green industrial policy. In my model, the “firm delocation” effect takes the form of a *green relocation effect*, by which an increase in the number of green firms in one country induces a reduction in the number of green firms in other countries. Second, I introduce a political economy process that shapes the incentives and constraints of policymakers. In a departure from the standard Grossman and Helpman (1994) approach, I assume that the *number* of firms determines the political power of any coalition of producers, rather than their willingness and ability to pay for policy concessions.¹ Through this channel, the green relocation effect has political consequences: policy that increases the number of green producers at home relaxing domestic

¹See, e.g., Lohmann (2023) for a critique of the Grossman-Helpman approach.

political constraints, but increases the political barriers for the implementation of climate policy abroad.

I find that in the optimal international agreement countries commit to free trade, a carbon tax equal to the global marginal cost of carbon emissions, and to green industrial policies if and only if domestic political constraints bind. If countries do not coordinate climate policies, in contrast, carbon taxes are too low, because policymakers do not internalize the global climate externality. The main result of the paper is that, in addition, governments can engage in subsidy races. The reason is that the level of the subsidy required to create a large enough coalition of green producers in one country increases with the level of green subsidies in other countries. In addition to relaxing political constraints, green subsidies have direct positive effects on economic welfare, which creates an addition incentive for their use. However, these effects are zero-sum, as they operate by relocating green firms from the foreign to the domestic economy. The second main result is that the subsidy race is not the only possible noncooperative outcome. Another possibility (which depends on parameters) is that one country pursues climate policy and, as a consequence, policy inaction results in other countries.

The most important conceptual takeaway of the analysis is that even though green industrial policy can be desirable due to (inter alia) its capacity to reduce domestic political opposition to more efficient climate policies, it can have negative *political* spillovers on climate mitigation in other countries. While existing research has recognized the usefulness of green subsidies to relax political constraints (e.g., [Hallegatte et al., 2024](#)), and their international economic spillovers, in particular the “reverse leakage” effect ([Kotchen and Maggi, 2025](#)), the point that uncoordinated green subsidies can be counterproductive has not been considered in previous work.

The immediate normative implication of the analysis is that countries should coordinate their use of green subsidies. A more subtle implication is that international trade law should allow for discriminatory green subsidies but should regulate them, since there are non-environmental incentives to use them. Existing international trade agreements incorporate provisions on environmental regulation, particularly about leakage ([Brandi and Morin, 2023](#)), but to the best of my knowledge no trade agreement has explicitly regulated the use of green subsidies. Reducing the damages from climate change will require both free trade ([Le Moigne et al., 2024](#)) and green industrial policies ([Hallegatte et al., 2024](#)), so the future will require their coordination.

Literature. The main contribution of the paper is to provide a model of green subsidy races, and study their implications for international cooperation, in particular for international trade law. [Clausing and Wolfram \(2023\)](#) and [Meckling \(2025\)](#) discuss subsidy races informally. Here I provide an explicit model that clarifies the forces that produce the subsidy race, namely, a combination of firm relocation and domestic political constraints. I show that a subsidy race is not the only possible outcome: without international cooperation, subsidies in one country may simply deter

other countries from pursuing climate policies, leading to a pattern of international specialization consistent with the argument in [Kennard and Melnick \(2025\)](#).

There is a large literature in economics and political science on international climate agreements, some of which studies the linkage of trade and climate policy (e.g., [Nordhaus, 2015](#); [Farrokhi and Lashkaripour, 2025](#)). The only papers that study international cooperation on both carbon taxes and green subsidies with international trade are [Kotchen and Maggi \(2025\)](#) and [Kay \(2025\)](#). In their models green subsidies are valuable because they have a “reverse leakage” effect: a subsidy on green goods reduces world prices, including that of polluting goods, which reduces their production. Both papers find that optimal international agreements have no green subsidies (only pollution taxes), but optimal noncooperative policies involve green subsidies. In their models there is no mechanism that generates a subsidy race. In [Kotchen and Maggi \(2025\)](#) lobbying by green producers induces policymakers to increase green subsidies (due to a simple quid-pro-quo, and not because green subsidies relax political constraints), and can improve welfare in the noncooperative equilibrium simply because it leads to more emissions abatement. The difference with my model is that I consider market imperfections that create a non-environmental reason to use green subsidies, which is consistent with empirical studies on the effects of subsidies ([Allcott et al., 2024](#)). The reverse leakage effect is not dominant in my model; in fact, green subsidies in one country can increase carbon emissions in other countries due to the green relocation effect.

There is a small literature on green protectionism and to what extent it conflicts with WTO rules, e.g., [Lewis \(2014\)](#), that I should summarize.

The idea that industrial policy more broadly has international spillovers and may require coordination in order to be effective is not new: see [Lashkaripour and Wu \(2025\)](#).

There is also a broader literature about the international coordination of different kinds of policies, e.g., [Tabellini \(1990\)](#) on the coordination of fiscal policy. There are some relevant ideas there, but nothing *super* relevant. There is, of course, an old literature on export promotion (possibly leading to subsidy races) started by [Brander and Spencer \(1985\)](#), but my understanding is that it failed to provide clear policy prescriptions (besides “don’t do it”) and at some point trade economics simply moved on. There is a modern literature on industrial policy and trade that is relevant. [Lashkaripour and Lugovskyy \(2023\)](#) find that unilateral industrial policy is counterproductive, but coordinated policies can bring welfare gains. [Bartelme et al. \(2025\)](#) find that industrial policy is ineffective, even though there are large external economies of scale in several sectors that (individually) justify these policies. The implications for *green* industrial policy specifically are not evident in my reading.

Comment. I’ll add more citations in the future (including all the political science I didn’t cite nor discuss, sorry!), and I’ll add more on the facts that motivate the question. Important questions that I plan to address include what is the best international agreement that can be enforced using

WTO procedures as in [Bagwell and Staiger \(2001\)](#) or a “climate club” that includes subsidies as threats. I may extend the analysis to a Melitz-style model (my model is a classical “new trade theory” model; this would upgrade it to a “new new trade theory” model).

2. The Model

2.1. Preface

I develop a model with the following features:

- There are two countries (Home and Foreign).
- There is a fixed set of manufacturing firms, each of which produces a distinct variety. Producers are monopolistic. Consumers value variety.
- There are two technologies available in each location: a polluting and a green technology.
- The green technology has a fixed cost, which implies increasing returns to scale.
- There is international trade with iceberg transport costs.
- Governments have access to carbon taxes, green industrial subsidies, and a full set of trade policy instruments (firm- and destination-specific tariffs and production subsidies).
- Governments maximize aggregate welfare, but policy changes require the approval of a certain number of manufacturing forms. (This feature creates political constraints.)

The combination of love of variety, increasing returns to scale, free entry, and transport costs creates a welfare rationale for the use of unilateral tariffs due to their “firm relocation” effect ([Venables, 1987](#); [Helpman and Krugman, 1989](#); [Ossa, 2011](#); [Bagwell and Staiger, 2015](#)). The intuition is as follows. Unilaterally increasing the cost of imports reduces demand for foreign varieties, which increases demand for domestic varieties. This increase in demand encourages the marginal domestic producer to enter the market. Therefore, more varieties are produced at home, which saves consumers the transport costs and increases aggregate welfare. However, this beneficial effect for the domestic economy comes at the expense of the foreign economy. Reducing domestic demand for foreign varieties induces the marginal foreign firm to exit the market. This is interpreted as the “relocation” effect: we can think of the marginal foreign producer as relocating to the home country in response to the tariff. Foreign consumers now have to pay transport costs for more products, which reduces their welfare. Because this incentive to raise tariffs is symmetric and negative-sum, both countries have a reason to sign a trade agreement in order to eliminate it. The optimal trade agreement involves no tariffs.

In my model, which features a fixed number of varieties and a technology choice that involves a fixed cost, asymmetric barriers to trade produce what we may call a *green relocation* effect. Creating a barrier for imports increases demand for domestic varieties. This, in turn, induces the marginal polluting firm to adopt the green technology, because doing this reduces marginal costs

(due to technological differences or to policy, e.g., a carbon tax). Thus, more firms become green in the domestic economy, and they increase their exports due to their lower marginal cost. This, in turn, increases competition abroad, which leads the marginal green firm to become unprofitable. If foreign firms anticipate the policy, the marginal firm planning to adopt the green technology will decide to keep using the polluting technology instead. The implication of this argument is that asymmetric trade restrictions can reduce emissions at home at the expense of increasing emissions abroad.

The key insight is that domestic climate policy also produces this green relocation effect. Reducing carbon emissions requires making the green technology more profitable than the polluting technology. This leads to an increase in the number of firms using the green technology at home. Assuming, again, that these firms have lower marginal costs, increasing their number creates more competition for firms abroad, which reduces their profitability, which leads some potentially green firms to stay polluting. We can interpret this shift as green investments being relocated from the foreign to the domestic economy.

2.2. Model of the Economy

There are two countries, H and F (Home and Foreign), a continuum of varieties $v \in [0, 1]$ of a manufacturing good per country, and a numéraire good N , which is freely and costlessly traded. Varieties face an iceberg trade cost $\theta > 1$, so that shipping a quantity q from i to j requires producing θq units of the good (the iceberg metaphor captures the assumption that a fraction $\theta - 1$ of the good “melts away” during transport; this is standard in trade economics). Consumer preferences in country j are given by

$$U_j = \log(Q_j) + N_j - D_j,$$

where

$$Q_j = \left(\sum_{i=H,F} \int_0^1 q_{ij}(v)^{1/\mu} dv \right)^\mu,$$

$q_{ij}(v)$ the quantity of variety v from i consumed in j , $\mu = \frac{\sigma}{\sigma-1}$, $\sigma > 1$ is the (constant) elasticity of substitution (CES) of varieties, N_j is the quantity of the numéraire consumed in j , and D_j captures climate damages, defined shortly.

Country i is endowed with labor L_i . I assume that L_i is large enough so that the solution to the consumer problem is always interior. Technology is the same in both countries. Each variety $v \in [0, 1]$ can be produced with a green or a polluting technology. (I will sometimes call them “clean” and “dirty.”) The green technology has labor cost $l = f + cq$ to produce q units, where $f > 0$ is a fixed cost and $0 < c < 1$ is the marginal cost, and the polluting technology has labor cost $l = q$. Producing q units of the numéraire requires labor $l = q$. Producers of varieties are

monopolists, and the numéraire good is produced competitively. Climate damages are

$$D_i \equiv \gamma \sum_{j=H,F} \int_0^1 (1 - \chi_j(v)) q_j(v) dv,$$

where $\chi_j(v) \in \{0, 1\}$ is 1 iff country j produces v with the green technology, $q_j(v) = q_{jj}(v) + \theta q_{jk}(v)$ (with $k \in \{H, F\}$, $k \neq j$) is the number of units of variety v produced at j , and $\gamma \geq 0$ measures the social cost of carbon. I am implicitly assuming that the carbon intensity of the polluting technology is 1 (i.e., producing one unit releases one unit of carbon pollution), and the carbon intensity of the clean technology is 0. Let $g_i \equiv \int_0^1 \chi_i(v) dv$ be the share of varieties produced with the green technology. Throughout the paper I will assume that in equilibrium $0 < g_i < 1$. [I should say: I will make the necessary assumptions to ensure that this is the case in every relevant scenario.]

I will consider the following policy instruments. Country i has two climate policies: it can impose a carbon tax $\tau_i \geq 0$ paid per unit of manufacturing varieties produced using the polluting technology, and a green subsidy $\phi_i \geq 0$ paid to producers that choose the green technology. In addition, countries have a complete set of trade policy instruments: country i can charge an ad-valorem tariff $t_{ji}(v)$ to consumption of variety v produced in country j , and provide an ad-valorem subsidy $s_{ij}(v)$ to production of variety v sold to country j . The fiscal revenue (or cost) of these policies is returned to consumers in the form of a uniform lump-sum transfer T_i (possibly negative).

The numéraire good is produced competitively and freely traded, so the wage is 1 in both countries. Varieties are produced by monopolists, who (by profit maximization) set ex-factory prices $\bar{p}_{ij}(v) \equiv \mu c_{ij}(v) / (1 + s_{ij}(v))$, where $\mu > 1$ is a markup (implied by CES demand), and $c_{ij}(v)$ is the marginal cost, including the transport cost θ in the case of exports and the carbon tax τ_i in the case of dirty varieties. Thus, green varieties have a marginal cost $c_{ij}(g) \equiv \theta^{1_{i \neq j}} c$, and polluting varieties have a marginal cost $c_{ij}(b) \equiv \theta^{1_{i \neq j}} (1 + \tau_i)$. Profits are $\pi_i(v) \equiv (\mu - 1) \sum_{j=H,F} c_{ij}(v) q_{ij}(v) - f_i(v)$, where $f_i(v) \equiv \chi_i(v)(f - \phi_i)$ is the fixed cost net of the green subsidy if v is produced using the green technology, and 0 if v is produced using the polluting technology. Labor costs are $l_i(v) \equiv [1 - \chi_i(v)(1 - c)] q_i(v) + \chi_i(v) f$.

Consumers face prices $p_{ij}(v) = (1 + t_{ij}(v)) \bar{p}_{ij}(v)$. Therefore, solving the consumer problem we obtain $q_{ij}(v) = p_{ij}(v)^{-\sigma} P_j^{\sigma-1}$, where $P_j = \left(\sum_{i=H,F} \int_0^1 p_{ij}(v)^{1-\sigma} dv \right)^{\frac{1}{1-\sigma}}$ is the “price index” of manufacturing varieties in country j , and $Q_j = P_j^{-1}$.

The indirect utility of the representative consumer is given by

$$W_i \equiv -\log P_i + L_i + \int_0^1 \pi_i(v) dv + T_i - 1 - D_i,$$

where

$$T_i = \tau_i(1 - g_i)q_i(b) - g_i\phi_i + \sum_{j=H,F} \int_0^1 t_{ji}\bar{p}_{ji}(v)q_{ji}(v) dv - \sum_{j=H,F} \int_0^1 s_{ij}\bar{p}_{ij}(v)q_{ij}(v) dv$$

is the lump-sum transfer. The first term is the revenue from the carbon tax, the second term is the fiscal cost of the green subsidy, the third term is the revenue from tariffs, and the last term is the fiscal cost of the production subsidies. An equivalent expression is

$$W_i = -\log P_i - \int_0^1 l_i(v) dv + \int_0^1 \bar{p}_{ij}(v)q_{ij}(v) dv - \int_0^1 \bar{p}_{ji}(v)q_{ji}(v) dv - D_i$$

plus a constant, where $j \in \{H, F\}$, $j \neq i$. The first term is the utility from consuming manufacturing products, the second term is the labor costs of manufacturing production in country i , the third term captures the revenue from exports, and the fourth term captures the costs of imports.

2.3. Model of the Policy Process

The status quo involves no climate policy in either country. I will specify the status quo trade policies in the analysis; the baseline assumption is that trade policies are exogenously given. The policymaker in country i can change climate policies but requires the support of a fraction $n_i \in [0, 1]$ of manufacturing producers to do so. Timing of events:

1. Policymakers simultaneously choose green subsidies $\phi_i \geq 0$ and carbon taxes $\tau_i \geq 0$.
2. Producers simultaneously choose technology.
3. Production and consumption takes place.

The policymaker seeks to maximize aggregate utility W_i , and producers' payoffs are given by their profits $\pi_i(v)$. The equilibrium concept is subgame perfect equilibrium.

2.4. Comments on the Assumptions

The assumption that adopting green technology requires a fixed cost and lowers marginal costs is consistent with the findings by [Colmer et al. \(2024\)](#), who show that emissions reductions after the EU ETS had no detectable contraction in economic activity, and interpret this pattern as consistent with firms making targeted fixed-cost investments in cleaner production technologies that reduced marginal variable costs.

The model of the economy is closely based on [Helpman and Krugman \(1989, Section 7.4\)](#), which was more recently revisited by [Bagwell and Staiger \(2015\)](#). The difference is that producers adjust not by entry decisions but by technological choices, and these choices involve a negative global externality (carbon emissions). I will also consider a full set of tax policy instruments, which

is something that neither treatment does. (I think I will also be able to analyze an extension with firm heterogeneity a la [Melitz, 2003](#).)

The assumption that policy changes require the support of a fraction n_i of manufacturing producers can be interpreted as follows. Each producer $v \in [0, 1]$ has a seat in a legislature in which the policymaker is the agenda setter, and the voting rule is such that any coalition containing at least n_i manufacturing producers is winning. The substantive meaning of this assumption is that producers can collectively block policy proposals, their power is given by numbers (not by, e.g., money), and a political actor who cares about aggregate welfare has full agenda control. This is the case if producers are located in different geographic locations, and these locations are politically represented by legislators whose electoral incentives are tied to the welfare of their district's producers. It is also the case if policy is negotiated between the government and a business association (as in corporatist systems) in which each producer has equal representation. My approach to modeling the power of the industry differs from what is standard in the literature, namely the common agency approach popularized by [Grossman and Helpman \(1994\)](#). For a critique of this approach see [Lohmann \(2023\)](#). The benefit of my approach is that it captures the idea that industrial policy can have political effects by increasing the size of the targeted industry.

I'm assuming that the economies are exactly symmetrical. I think I need *some* symmetry, because otherwise I can't have an interior number of green varieties in both countries, but I don't need *full* symmetry. I'll relax this assumption in a future iteration, possibly in an appendix if it only complicates things.

3. Climate Policy in Autarky

Before I analyze the full model, let's study a version of the model with only one country. We can drop subscripts.

3.1. No Political Constraints

Without political constraints, optimal policy consists of carbon taxes equal to marginal climate damage, $\tau = \gamma$, no green subsidies, $\phi = 0$, and a uniform production subsidy $s = \mu - 1$. This policy choice is time-consistent (i.e., it doesn't depend on whether the policymaker can commit to it or not), because the optimal carbon tax does not depend on the number of firms using green technology. The production subsidy is required to correct for the distortion that results from monopoly power.

We can prove this standard result by noting that $\tau = \gamma$, $\phi = 0$, $t = 0$ and $s = \mu - 1$ in fact implement the optimal allocation. We can use the primal approach, namely, we can allow the policymaker to choose quantities and technologies directly (as a social planner), and then show that the optimal allocation is in fact implemented by those policies. The planner's problem is to choose $(q(v), \chi(v))_{v \in [0,1]}$ and Q, N to maximize welfare $W = \log Q + N - D$ subject to

$Q^{1/\mu} = \int_0^1 q(v)^{1/\mu} dv$ and the resource constraint $\int_0^1 l(v) dv + N \leq L$. After eliminating N , the Lagrangian is

$$\begin{aligned} \mathcal{L} = & \log Q - \int_0^1 [\chi(v)(cq(v) + f) + (1 - \chi(v))q(v)] dv - \gamma \int_0^1 (1 - \chi(v))q(v) dv \\ & - \lambda \left(Q^{1/\mu} - \int_0^1 q(v)^{1/\mu} dv \right) \end{aligned}$$

for some $\lambda \in \mathbb{R}$. The FOC for Q is $\lambda = \mu Q^{-1/\mu}$. Replacing, the FOC for $q(v)$ is

$$q(v) = [\chi(v)c + (1 - \chi(v))(1 + \gamma)]^{-\sigma} Q^{1-\sigma}. \quad (*)$$

We can sort the varieties so that $\chi(v) = 1_{v \leq g}$ for some $g \in [0, 1]$, and we get

$$\begin{aligned} \frac{d\mathcal{L}}{dg} &= (1 + \gamma)q(1) - cq(g) - f + \mu Q^{-1/\mu} [q(g)^{1/\mu} - q(1)^{1/\mu}] \\ &= (\mu - 1)cq(g) - f - (\mu - 1)(1 + \gamma)q(1). \end{aligned}$$

Therefore an interior q requires

$$(\mu - 1)cq(g) - f = (\mu - 1)(1 + \gamma)q(1). \quad (**)$$

If $\tau = \gamma$, $\phi = 0$, $t = 0$ and $s = \mu - 1$ then we have $p(v) = \chi(v)c + (1 - \chi(v))(1 + \gamma)$, so demand is $q(v) = p(v)^{-\sigma} P^{\sigma-1} = [\chi(v)c + (1 - \chi(v))(1 + \gamma)]^{-\sigma} Q^{1-\sigma}$, i.e., (*), and an interior g obtains if producers are indifferent between the two technologies, which happens iff $\pi(g) = \pi(b)$, i.e., iff (**). In conclusion, the policies implement the optimal allocation, and are, therefore, optimal, as claimed.

What this result shows is that in the model the optimal climate policy consists of a Pigouvian carbon tax (i.e., one that makes producers pay for their marginal social cost); the green investment subsidy is not needed. This is because the Pigouvian tax provides the right incentives both for the level of consumption of dirty varieties (those produced with the polluting technology), and for technology choices (how many firms should adopt the clean technology).

3.2. Political Constraints

Profits using the polluting technology are decreasing in the carbon tax:

$$\pi(b) = (\mu - 1)(1 + \tau)q(b) = (\mu - 1)(1 + \tau)p(b)^{-\sigma} P^{\sigma-1} = \frac{(\mu - 1)(1 + \tau)p(b)^{-\sigma}}{gp(g)^{1-\sigma} + (1 - g)p(b)^{1-\sigma}}$$

with $p(b) = \mu(1 + \tau)(1 + t)/(1 + s)$ and $p(g) = \mu c/(1 + s)$, so clearly $\frac{d\pi(b)}{d\tau} < 0$. Therefore, polluters refuse to accept an increase in carbon taxes. Producers who adopted the green technology, in contrast, benefit from an increase in the carbon tax, because it reduces their competition from polluting producers and increases their profits.

Therefore, if the number of green producers in the optimal allocation is smaller than the size of the minimal coalition required to implement a policy change ($g^* < n$), the optimal policy cannot be implemented. Now, the green subsidy increases the number of green producers, and therefore in equilibrium the policymaker needs to use the subsidy so that $g = n$, and then set $\tau = \gamma$ [Oh, I haven't proved the latter, but it's obvious, trust me]. If this policy increases aggregate welfare relative to the status quo, it will be acceptable to the coalition of producers who decide to adopt the green technology [I have to prove this, but I believe it's also obvious].

In other words, political constraints may force policymakers to use green subsidies, even though there isn't a welfare rationale for the subsidies in the model.

4. Climate Policy with International Spillovers

I'm going to start by looking for the optimal policy chosen by a global planner without political constraints. This benchmark shows that trade policy and green subsidies are not good policy instruments in the model, and therefore their use can only be explained by failures of international cooperation or by domestic political constraints. [This obviously shouldn't be taken literally as a normative claim. It just means that the model is designed to focus on those reasons for the use of those policies.] Next I introduce political constraints, but keep the possibility of an international agreement on trade and climate policies. I find that the green subsidy is *the* right policy instrument for accommodating domestic political constraints in an optimal international agreement. Next I drop the possibility of an international agreement, but this time I keep the assumption of no political constraints. I find that the optimal unilateral (or Nash) trade-plus-climate policy uses tariffs and export subsidies or taxes, that the optimal tariffs include a carbon tax but the export subsidies do not, and that the green subsidy is not used. The takeaway is that green subsidies are not the right policy instrument for exploiting international externalities. Green subsidies are a first-best policy instrument for dealing with domestic political constraints, but they are (at best) a second-best trade policy instrument.

I then study the actual model, assuming that trade policies are fixed before the game starts by an optimal trade agreement that was designed without climate objectives. We know that this agreement involves uniform non-discriminatory production subsidies that undo the monopoly distortion, and no import tariffs. Here I obtain the three main results. The first is that green subsidies chosen unilaterally are in fact useful as second-best trade policies because of their "green relocation" effect. The second result is that, if domestic political constraints are not binding, Nash green subsidies are

larger than those chosen if only one country can use them. This captures the idea of a subsidy race. The third result is that if domestic political constraints are more binding in one country, the other country can choose their green subsidies strategically to make them bind and therefore stop the subsidy race. In other words, a country may use green subsidies to “weaponize” domestic political constraints in other countries. The unfortunate implication is that this strategy prevents the target country from using *any* climate policy (including carbon taxes), which is bad for global welfare.

4.1. *Benchmark: International Cooperation without Political Constraints*

Suppose that policymakers in both countries can choose policies without political constraints, and coordinate their policies in a negotiated international agreement. The same argument in [Subsection 3.1](#) can be used to show that the optimal allocation can be implemented setting $\tau_i = 2\gamma$, $\phi_i = 0$, no tariffs, and uniform non-discriminatory production subsidies $s_{ij} = \mu - 1$.

International cooperation without domestic political constraints involves more ambitious climate change mitigation policies (because countries agree to internalize the externality they impose on each other) and free trade. Green subsidies are not needed: with the right carbon price, producers make the socially optimal choices regarding pollution abatement.

However, domestic political constraints can prevent policymakers from implementing their policy commitments. If the equilibrium number of green firms given $\tau_i = 2\gamma$ and $\phi_i = 0$ is smaller than the size of the minimal coalition, policymakers will not be able to ratify the agreement at home.

4.2. *International Cooperation with Binding Political Constraints*

What is the best international agreement subject to domestic political constraints? I can use again the primal approach. The planner’s problem is to maximize

$$\sum_{i=H,F} W_i = \sum_{i=H,F} (\log Q_i + N_i - D_i)$$

subject to

$$Q_i^{1/\mu} = \sum_{j=H,F} \int_0^1 q_{ji}(v)^{1/\mu} dv,$$

the aggregate resource constraint

$$\sum_{i=H,F} \left\{ \int_0^1 l_i(v) dv + N_i \right\} = \sum_{i=H,F} L_i,$$

and the domestic political constraints $g_i \geq n_i$.

The Langrangian, after eliminating N_i , is

$$\begin{aligned}\mathcal{L} = & \sum_{i=H,F} \left\{ \log Q_i - \int_0^1 l_i(v) dv - 2\gamma \int_0^1 (1 - \chi_i(v)) q_i(v) dv \right\} \\ & - \sum_{i=H,F} \lambda_i \left(Q_i^{1/\mu} - \sum_{j=H,F} \int_0^1 q_{ji}(v)^{1/\mu} dv \right) + \sum_{i=H,F} \eta_i (g_i - n_i),\end{aligned}$$

with $\lambda_i \in \mathbb{R}$, $\eta_i \geq 0$ and $\eta_i (g_i - n_i) = 0$. Using exactly the same approach as in [Subsection 3.1](#) we obtain that the optimal allocation can be implemented by using carbon taxes $\tau_i = 2\gamma$, green subsidies $\phi_i = \eta_i$, no tariffs, and uniform non-discriminatory production subsidies $s_{ij} = \mu - 1$.

In sum, the optimal international agreement subject to domestic political constraints involves carbon taxes equal to the (global) social cost of carbon, free trade, and green subsidies if and only if political constraints are binding (i.e., $\eta_i > 0$).

4.3. Unilateral Trade and Climate Policies

Now I'm going to study what happens if each country can choose trade and climate policies unilaterally, without domestic political constraints (this includes commitment power vis-a-vis domestic producers).

We can always order varieties so that $\chi_i(v) = 1_{v \leq g_i}$ for some $g_i \in [0, 1]$. I'm going to drop the dependence on v to ease notation, but don't forget it's there.

I use the primal approach, following [Costinot et al. \(2020\)](#). Home's planner takes trade $(t_{FF}, t_{HF}, s_{FF}, s_{FH})$ and climate policies (τ_F, ϕ_F) in Foreign as given. The planner can choose quantities consumed at Home, q_{HH}, q_{FH} , technology choices g_H , and ex-factory export prices $\bar{p}_{HF}$.

Endogenous variables are determined as follows. Ex-factory prices at F are $\bar{p}_{FF} = \mu c_F / (1 + s_{FF})$ and $\bar{p}_{FH} = \mu c_F / (1 + s_{FH})$, where $c_F \equiv \chi_F c + (1 - \chi_F)(1 + \tau_F)$. The quantities demanded at F are determined in equilibrium by $q_{FF} = \bar{p}_{FF}^{-\sigma} P_F^{\sigma-1}$ and $q_{HF} = [(1 + t_{FH}) \bar{p}_{HF}]^{-\sigma} P_F^{\sigma-1}$, where $p_{FF} = (1 + t_{FF}) \bar{p}_{FF}$ and $P_F^{1-\sigma} = \int p_{FF}^{1-\sigma} + \int [(1 + t_{FF}) \bar{p}_{HF}]^{1-\sigma}$. The number of green varieties g_F is determined by the iso-profit condition $\pi_F(b) = \pi_F(g)$, where $b \equiv 1$ is any dirty variety. Let $f_F \equiv \chi_F (f - \phi_F)$, and $c_H \equiv \chi_H c + (1 - \chi_H)(1 + \gamma)$.

We can write the planner's problem as follows. The planner chooses $q_{HH}, q_{FH}, g_H, \bar{p}_{HF}, Q_H, P_F, g_F$ to maximize

$$\begin{aligned}W_H = & \log Q_H - \int [c_H(q_{HH} + \theta q_{HF}) + \chi_H f] + \int \bar{p}_{HF} q_{HF} - \int \bar{p}_{FH} q_{FH} \\ & - \gamma \int (1 - \chi)(q_{FF} + \theta q_{FH})\end{aligned}$$

subject to

$$\begin{aligned} Q_H^{1/\mu} &= \int q_{HH}^{1/\mu} + \int q_{FH}^{1/\mu}, \\ P_F^{1-\sigma} &= \int p_{FF}^{1-\sigma} + \int [(1 + t_{HF})\bar{p}_{HF}]^{1-\sigma}, \\ \Delta[(\mu - 1)c_F(q_{FF} + \theta q_{FH}) - f_F] &= 0, \end{aligned}$$

where the $\Delta[\cdot]$ operator means $\Delta[f] \equiv f(b) - f(g)$, and the last constraint is simply $\pi_F(b) = \pi_F(g)$. The Lagrangian is

$$\begin{aligned} \mathcal{L}_H &= \log Q_H - \int [c_H(q_{HH} + \theta q_{HF}) + \chi_H f] + \int \bar{p}_{HF} q_{HF} - \int \bar{p}_{FH} q_{FH} \\ &\quad - \gamma \int (1 - \chi)(q_{FF} + \theta q_{FH}) - \lambda_H \left(Q_H^{1/\mu} - \int q_{HH}^{1/\mu} - \int q_{FH}^{1/\mu} \right) \\ &\quad - \lambda_F \left(P_F^{1-\sigma} - \int p_{FF}^{1-\sigma} - \int [(1 + t_{HF})\bar{p}_{HF}]^{1-\sigma} \right) + \eta_F \Delta[(\mu - 1)c_F(q_{FF} + \theta q_{FH}) - f_F], \end{aligned}$$

for some $\lambda_H, \lambda_F, \eta_F \in \mathbb{R}$, where

$$\begin{aligned} p_{FF} &= (1 + t_{FF})\mu c_F / (1 + s_{FF}), \\ q_{FF} &= p_{FF}^{-\sigma} P_F^{\sigma-1}, \text{ and} \\ q_{HF} &= [(1 + t_{HF})\bar{p}_{HF}]^{-\sigma} P_F^{\sigma-1}. \end{aligned}$$

We can write the first-order conditions as follows [full derivation in an appendix I'll write eventually]:

$$\begin{aligned} q_{HH} &= c_H^{-\sigma} Q_H^{1-\sigma}, \\ q_{FH} &= \begin{cases} \left[\left(1 + \left(\frac{\gamma}{\mu} - \frac{\eta_F}{\sigma(1-g_F)} \right) (1 + s_{FH}) \right) \bar{p}_{FH}(b) \right]^{-\sigma} Q_H^{1-\sigma} & \text{if } \chi = 0, \\ \left[\left(1 + \frac{\eta_F}{\sigma g_F} (1 + s_{FH}) \right) \bar{p}_{FH}(g) \right]^{-\sigma} Q_H^{1-\sigma} & \text{if } \chi = 1, \end{cases} \\ \bar{p}_{HF} &= \mu \theta c_H / (1 + \lambda_F (1 + t_{HF}) P_F^{1-\sigma}), \quad \text{and} \\ \Delta[(\mu - 1)c_H(q_{HH} + \theta q_{HF}) - f_H] &= 0, \end{aligned}$$

where $f_H = \chi_H f$.

We can implement this allocation by choosing an import tariff that is $t_{FH} = \left(\frac{\gamma}{\mu} - \frac{\eta_F}{\sigma(1-g_F)} \right) (1 + s_{FH})$ for dirty varieties and $t_{FH} = \frac{\eta_F}{\sigma g_F} (1 + s_{FH})$ for clean varieties, a production subsidy for the domestic market $s_{HH} = \mu - 1$, an export subsidy $s_{HF} = \lambda_F (1 + t_{HF}) P_F^{1-\sigma}$, a carbon tax $\tau_H = \gamma$, and no green subsidies.

There are several reasons for using trade policy in this model. First, correcting the monopoly distortion is useful for consumption at home, but not abroad. The planner is happy to apply the markup μ to exports, because the profits stay at home and the cost is paid by foreign consumers. Second, the planner uses export taxes or subsidies to exploit the aggregate market power of the industry as a whole, internalizing the effect on g_F . Third, the planner uses import tariffs on dirty products as a carbon tax. In other words, the planner implements a carbon border adjustment mechanism (CBAM), as the EU currently does. Finally, the planner uses import tariffs to change the incentives driving foreign producers' technology choices.

I conjecture that in the Nash equilibrium we have $\lambda_F > 0$ (i.e., there is an export subsidy) and $\eta_F > 0$ (i.e., imports of clean products are discouraged with a tariff, and imports of dirty products are encouraged by relaxing the CBAM). The reason for these distortions is that they reduce g_F and increase g_H , i.e., they exploit the green relocation effect.

The important takeaway is that with a full set of trade policy instruments, (1) the equilibrium carbon tax is equal to the *domestic* social cost of carbon (not the *global* social cost, which is the relevant quantity; this is the classic free-riding logic), and (2) green investment subsidies are not needed.

4.4. Green Subsidies and the Green Relocation Effect

We can analyze the political game now. I am going to assume that the countries signed an optimal trade agreement ignoring the climate externality before starting the game, and, therefore, there are no tariffs, and there are uniform non-discriminatory consumption subsidies $t_{ij} = -1/\sigma$ (for every $i, j \in \{H, F\}$, so that $(1 + t_{ij})\mu = 1$) used to remove the markup due to monopolistic competition and restore efficiency. Recall the timing:

1. Policymakers simultaneously choose green subsidies $\phi_i \geq 0$ and carbon taxes $\tau_i \geq 0$.
2. Producers simultaneously choose technology.
3. Production and consumption takes place.

Welfare at Home is

$$\begin{aligned}
W_H &= -\log P_H - g_H[cq_H(g) + f] - (1 - g_H)q_H(b) \\
&\quad + g_H\mu\theta cq_{HF}(g) + (1 - g_H)\mu\theta(1 + \tau_H)q_{HF}(b) \\
&\quad - g_F\mu\theta cq_{FH}(g) - (1 - g_F)\mu\theta(1 + \tau_F)q_{FH}(b) \\
&\quad - \gamma(1 - g_H)q_H(b) - \gamma(1 - g_F)q_F(b) \\
&= -\log P_H - g_H f + (1 - g_H)(\tau_H - \gamma)q_H(b) - (1 - g_F)\gamma q_F(b) - 1 \\
&\quad + (\mu - 1)[g_H\theta cq_{HF}(g) + (1 - g_H)\theta(1 + \tau_H)q_{HF}(b) \\
&\quad - g_F\theta cq_{FH}(g) - (1 - g_F)\theta(1 + \tau_F)q_{FH}(b)],
\end{aligned}$$

where

$$\begin{aligned}
P_i^{1-\sigma} &= g_i c^{1-\sigma} + (1 - g_i)(1 + \tau_i)^{1-\sigma} + g_j(\theta c)^{1-\sigma} + (1 - g_j)(\theta(1 + \tau_j))^{1-\sigma}, \\
q_i(t) &= q_{ii}(t) + \theta q_{ij}(t) \text{ for each } t = g, b, \\
q_{ii}(g) &= c^{-\sigma} P_i^{\sigma-1}, \\
q_{ii}(b) &= (1 + \tau_i)^{-\sigma} P_i^{\sigma-1}, \\
q_{ij}(g) &= (\theta c)^{-\sigma} P_j^{\sigma-1}, \\
q_{ij}(b) &= (\theta(1 + \tau_i))^{-\sigma} P_j^{\sigma-1}, \text{ and} \\
\Delta[(\mu - 1)c_i q_i - f_i] &= \phi_i
\end{aligned}$$

for every $i, j \in \{H, F\}$, $i \neq j$, where $c_i \equiv \chi_i c + \chi_i(1 + \tau_i)$ and $f_i \equiv \chi_i f$. We obtain W_F in a symmetric way. [Recall the assumptions: $\sigma > 1$, $\theta > 1$, $0 < c < 1$, $f > 0$, and $\gamma \geq 0$. Notation: $\mu \equiv \frac{\sigma}{\sigma-1}$, $\chi_i(t) \equiv 1_{t=g}$ for $t = g, b$, and $\Delta[F] \equiv F(b) - F(g)$.]

Suppose first that political constraints are not present. We can prove the following.

PROPOSITION 1. *Assume*

$$\frac{1 - \left(\frac{c}{1+\gamma}\right)^{\sigma-1}}{\sigma - 1} < f < \frac{\left(\frac{1+\gamma}{c}\right)^{\sigma-1} - 1}{\sigma - 1}$$

and

$$2\gamma(\sigma - 1) < (1 - \theta^{1-\sigma})(1 + \gamma) \left[\left(\frac{1 + \gamma}{c}\right)^{\sigma-1} - 1 \right].$$

(This is the case if γ is small, for example.) Then there exists $\bar{\theta} > 1$ such that, for every $\theta > \bar{\theta}$, there is a unique Nash equilibrium. It is symmetric, and $\tau_H = \tau_F < \gamma$, $\phi_H = \phi_F > 0$.

*Proof (sketch)*². When $\theta^{1-\sigma} = 0$ the two countries decouple, and for each country i ,

$$\frac{dW_i}{d\tau_i} = (\tau_i - \gamma)(1 - g_i) \underbrace{\frac{\partial q_i(b)}{\partial \tau_i}}_{<0}, \quad \frac{dW_i}{dg_i} = -\phi_i - (\tau_i - \gamma) \left[q_i(b) - (1 - g_i) \frac{\partial q_i(b)}{\partial g_i} \right].$$

Thus in equilibrium we have $\tau_i = \gamma$. The technology-indifference condition is

$$f = \frac{c^{1-\sigma} - (1 + \gamma)^{1-\sigma}}{(\sigma - 1)[g_i c^{1-\sigma} + (1 - g_i)(1 + \gamma)^{1-\sigma}]}.$$

²This was mostly written by ChatGPT. Obviously in the next version I'm going to write this properly, and I will complete the argument needed to show that the equilibrium is unique. But the idea and the computations are more or less there.

The right-hand side is strictly decreasing in g_i , and it takes the values

$$\frac{\left(\frac{1+\gamma}{c}\right)^{\sigma-1} - 1}{\sigma - 1} \quad \text{at } g_i = 0, \quad \frac{1 - \left(\frac{c}{1+\gamma}\right)^{\sigma-1}}{\sigma - 1} \quad \text{at } g_i = 1.$$

Hence the assumption on f is exactly the condition for a unique solution $g_i \in (0, 1)$.

At this decoupled point, all foreign derivatives vanish. Also,

$$\frac{\partial}{\partial g_i} \frac{dW_i}{d\tau_i} \Big|_{\tau_i=\gamma} = 0.$$

Therefore the Jacobian of the four first-order conditions with respect to $(\tau_H, g_H, \tau_F, g_F)$ is block lower triangular. Its diagonal entries are

$$-\frac{(1 - g_i)(1 + \gamma)^{-\sigma}}{(1 + \gamma) [g_i c^{1-\sigma} + (1 - g_i)(1 + \gamma)^{1-\sigma}]} \left[\sigma - (\sigma - 1) \frac{(1 - g_i)(1 + \gamma)^{1-\sigma}}{g_i c^{1-\sigma} + (1 - g_i)(1 + \gamma)^{1-\sigma}} \right] < 0$$

for the τ_i -equation and

$$-\frac{(c^{1-\sigma} - (1 + \gamma)^{1-\sigma})^2}{(\sigma - 1) [g_i c^{1-\sigma} + (1 - g_i)(1 + \gamma)^{1-\sigma}]^2} < 0$$

for the g_i -equation. Thus the Jacobian is nonsingular. The implicit-function theorem then gives $\bar{\theta} > 1$ such that for every $\theta > \bar{\theta}$ there is a unique interior solution of the first-order conditions in a neighborhood of the decoupled point. Since the game is symmetric, swapping H and F produces another local solution; by local uniqueness, the local solution must be symmetric.

Now fix any symmetric interior solution and write it as

$$\tau_H = \tau_F = \tau, \quad \phi_H = \phi_F = \phi, \quad g_H = g_F = g.$$

Then

$$P_H = P_F = P, \quad P^{1-\sigma} = (1 + \theta^{1-\sigma}) [g c^{1-\sigma} + (1 - g)(1 + \tau)^{1-\sigma}],$$

$$q_H(b) = q_F(b) = \frac{(1 + \tau)^{-\sigma}}{g c^{1-\sigma} + (1 - g)(1 + \tau)^{1-\sigma}},$$

$$\phi = f - \frac{c^{1-\sigma} - (1 + \tau)^{1-\sigma}}{(\sigma - 1) [g c^{1-\sigma} + (1 - g)(1 + \tau)^{1-\sigma}]},$$

$$\frac{dg_F}{d\tau_H} = \frac{2(\sigma - 1)(1 - g)\theta^{1-\sigma}}{(1 + \tau)^\sigma (c^{1-\sigma} - (1 + \tau)^{1-\sigma}) (1 + \theta^{2-2\sigma})}, \quad \frac{dg_F}{dg_H} = -\frac{2\theta^{1-\sigma}}{1 + \theta^{2-2\sigma}}.$$

Substituting these expressions into the total derivatives gives

$$0 = -\frac{(1-g)q_H(b)}{1+\tau} \left[\sigma - (\sigma-1) \frac{(1-g)(1+\tau)^{1-\sigma}}{gc^{1-\sigma} + (1-g)(1+\tau)^{1-\sigma}} \frac{(1-\theta^{1-\sigma})^2}{1+\theta^{2-2\sigma}} \right] (\tau - \gamma) \\ - (1-g)q_H(b) \frac{\theta^{1-\sigma}(1-\theta^{1-\sigma})}{1+\theta^{2-2\sigma}} + \frac{2\gamma(\sigma-1)\theta^{1-\sigma}}{(1+\theta^{2-2\sigma})(1+\tau)} \frac{(1-g)q_H(b)}{\left(\frac{1+\tau}{c}\right)^{\sigma-1} - 1}, \quad (1)$$

and

$$0 = -\frac{1+\theta^{1-\sigma}}{1+\theta^{2-2\sigma}}\phi + \frac{\theta^{1-\sigma}(1-\theta^{1-\sigma})}{1+\theta^{2-2\sigma}}f \\ - (\tau - \gamma)q_H(b) \left[1 + (1-g) \frac{c^{1-\sigma} - (1+\tau)^{1-\sigma}}{gc^{1-\sigma} + (1-g)(1+\tau)^{1-\sigma}} \frac{(1-\theta^{1-\sigma})^2}{1+\theta^{2-2\sigma}} \right] \\ - \frac{2\gamma\theta^{1-\sigma}}{1+\theta^{2-2\sigma}}q_H(b). \quad (2)$$

Solving (1) for $\tau - \gamma$ gives

$$\tau - \gamma = \frac{\theta^{1-\sigma}}{1+\theta^{2-2\sigma}} \frac{\frac{2\gamma(\sigma-1)}{\left(\frac{1+\tau}{c}\right)^{\sigma-1} - 1} - (1-\theta^{1-\sigma})(1+\tau)}{\sigma - (\sigma-1) \frac{(1-g)(1+\tau)^{1-\sigma}}{gc^{1-\sigma} + (1-g)(1+\tau)^{1-\sigma}} \frac{(1-\theta^{1-\sigma})^2}{1+\theta^{2-2\sigma}}}.$$

Because

$$0 < \frac{(1-g)(1+\tau)^{1-\sigma}}{gc^{1-\sigma} + (1-g)(1+\tau)^{1-\sigma}} < 1 \quad \text{and} \quad 0 < \frac{(1-\theta^{1-\sigma})^2}{1+\theta^{2-2\sigma}} < 1,$$

the denominator is strictly positive. Hence $\tau - \gamma$ has the same sign as

$$\frac{2\gamma(\sigma-1)}{\left(\frac{1+\tau}{c}\right)^{\sigma-1} - 1} - (1-\theta^{1-\sigma})(1+\tau).$$

This expression is strictly decreasing in τ . Therefore $\tau < \gamma$ iff

$$2\gamma(\sigma-1) < (1-\theta^{1-\sigma})(1+\gamma) \left[\left(\frac{1+\gamma}{c}\right)^{\sigma-1} - 1 \right].$$

Next, the technology-indifference condition implies

$$f - \phi = \frac{c^{1-\sigma} - (1+\tau)^{1-\sigma}}{(\sigma-1) [gc^{1-\sigma} + (1-g)(1+\tau)^{1-\sigma}]} = \frac{(1+\tau)q_H(b)}{\sigma-1} \left[\left(\frac{1+\tau}{c}\right)^{\sigma-1} - 1 \right].$$

Using this identity in (2), and then using (1) to eliminate the terms involving $\theta^{1-\sigma}(1-\theta^{1-\sigma})$ and

γ , yields

$$\phi = -(\tau - \gamma)q_H(b) \left[1 + \mu \left(\left(\frac{1 + \tau}{c} \right)^{\sigma-1} - 1 \right) \right].$$

Hence $\phi > 0$ if $\tau < \gamma$. This proves the proposition. ■

When the political constraints $g_i \geq n_i$ don't bind, what the Proposition suggests is that the equilibrium policy is symmetric, and involves carbon taxes lower than the social cost of carbon γ , and positive subsidies. The intuition is that the subsidies are used for their green relocation effect. (As the proof shows, without trade the equilibrium policy is carbon taxes equal to the social cost of carbon and no green subsidies.) The carbon tax is then lower than γ , because the green subsidies are (inefficiently) doing some of the work needed to correct the climate externality. This is one of the main points of the paper.

The next question is how political constraints affect equilibrium policies. Suppose that n_F increases until the political constraint at Foreign $g_F \geq n_F$ binds. If n_F increases, g_F has to increase, keeping $g_F = n_F$, and $(\tau_H, \phi_H, \tau_F, \phi_F)$ have to respond. The policymaker at Home still chooses τ_H, ϕ_H to maximize W_H taking τ_F, ϕ_F as given, subject to $g_H \geq n_H$. But now the policymaker at Foreign has to choose τ_F, ϕ_F to maximize W_F subject to $g_F = n_F$. What I'm in the process of proving is that as n_F increases a little bit, τ_H and g_H decrease, but ϕ_H increases. In other words, if political constraints at Foreign are tighter, climate policy at Home becomes weaker. It seems that W_F goes down, which is intuitive, because F is constrained, but W_H goes up.

If we extrapolate, as n_F increases, at some point either $g_H \geq n_H$ binds, or the policymaker at Foreign prefers not to implement any climate policy (i.e., the constraint $W_F \geq \underline{W}_F$ binds, where $\underline{W}_F = W_F|_{\tau_F=\phi_F=0}$). What I conjecture is that at this point there is no pure strategy equilibrium, because if Home expects $\tau_F = \phi_F = 0$, they will choose τ_H, ϕ_H to maximize W_H subject to $\tau_F = \phi_F = 0$, which requires a much smaller subsidy ϕ_H ; but because ϕ_H is smaller, Foreign can achieve $g_F = n_F$ with a smaller ϕ_F , so with this choice $W_F > \underline{W}_F$, which contradicts that $\tau_F = \phi_F = 0$ is an equilibrium. What Home can do is to choose ϕ_H larger than the best response to $\tau_F = \phi_F = 0$ with some probability in order to try to deter Foreign from using climate policy.

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